

Jay William Love, PhD

Postdoctoral Research Associate

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Primary Research Interests

- Evolution of collective behavior
- Ecology and evolution of communication systems
- Evolutionary interactions between infectious disease and sociality
- Role of collectives in ecology, ecological resilience to disturbance
- Application of machine learning tools in conservation biology

Education, Training, and Academic Positions

2025.1 – Present	Cornell University , College of Agriculture and Life Science <i>Postdoctoral Research Associate</i> Department of Computational Biology Advisor: Andrew Hein
2022.4 – 2025.1	University of Utah , School of Medicine <i>Research Associate</i> Division of Epidemiology Supervised by: Damon Toth, Matt Samore
2022.4 – 2025.1	US Centers for Disease Control and Prevention , National Center for Emerging and Zoonotic Infectious Diseases <i>Guest Researcher</i> Division of Healthcare Quality Promotion Supervised by: Rachel Slayton
2020.10 – 2022.4	University of Utah , School of Medicine <i>Postdoctoral Fellow</i> Division of Epidemiology Advisors: Damon Toth, Matt Samore
2019.8 – 2020.10	University of Utah , School of Medicine <i>Postdoctoral Fellow</i> Department of Neurobiology and Anatomy Advisor: Moriel Zelikowsky
2014.8 – 2019.8	University of Utah , School of Biological Sciences <u><i>PhD Biology</i></u> "Implications of Modular Song Learning for Species Isolation" Advisor: Franz Goller
2007.8 – 2010.12	University of Georgia , Eugene P. Odum School of Ecology <u><i>B.S. Ecology</i></u>

Publications

* = mentee

In Press/Submitted/Preprint

- **Love J**, Nayak A, Grassick A, Henry E, Kong LW, Fahimipour A, Gil M, Pennell M, Hein A (*in review following revision at PNAS*) Evolutionary dynamics of collective behavior across the tree of ray-finned fishes.
- See preprint: doi.org/10.64898/2026.04.06.716557
- **Love J**, Cooper BF, Plamondon S, Goller F (*in press at Journal of Comparative Physiology A*) Motor control of song and song-like vocalizations in the white-crowned sparrow.
- Nguyen D, Duong KNC, Coates E, Nelson RE, **Love J**, Jones M, Samore M, Chaiyakunapruk N, Toth D (*submitted to BMC Public Health*) Assessment of the effects of an equity-targeted vaccination outreach program on overall and distributional health and economic outcomes
- **Love J**, LaPrete CR*, Sheets TR, Vega Yon GG, Thomas A, Samore MH, Keegan LT, Adler FR, Slayton R, Spicknall I, Toth DJ (*in revision/preprint*) Characterizing spatiotemporal variation in transmission heterogeneity during the 2022 mpox outbreak in the USA. doi:10.1101/2023.05.10.23289580

Published

- Kong LW, Gallart L, Grassick AG, **Love J**, Nayak A, Hein AM (2025) A brief natural history of misinformation. *Journal of the Royal Society Interface* 22 (233): 20250161. doi: 10.1098/rsif.2025.0161
- See press: NY Times, Carl Zimmer:
<https://www.nytimes.com/2025/12/09/science/evolution-fish-misinformation.html>
- Toth DJA, Sheets TR, Beams AB, Ahmed SM, Seegert N, **Love J**, Keegan LT, Samore MH (2024) Model-based estimates of age-structured SARS-CoV-2 epidemiology in households. *BMC Public Health* 24(2965) doi: 10.1101/2024.04.18.24306047
- Duong KNC, Nguyen DT, Kategeaw W, Liang X, Khaing W, Visnovsky LD, Veetil SK, McFarland MM, Nelson RE, Jones BE, Pavia AT, Coates E, Khader K, **Love J**, Vega Yon GG, Zhang Y, Willson T, Dorsan E, Toth DJA, Jones MM, Samore MH, Chaiyakunapruk N (2024) Incorporating social determinants of health into transmission modeling of COVID-19 vaccine in the US: a scoping review. *Lancet Regional Health – Americas* 35(100806) doi:10.1016/j.lana.2024.100806
- Longtine C, Eliason CM, Lee CH, Mishkind D, Chiappone M, Goller F, **Love J**, Kingsley EP, Clarke JA, Tabin CJ. (2024) Homology and the evolution of vocal folds in the novel avian voice box. *Current Biology* 34(3): 461-472. e7. doi:10.1016/j.cub.2023.12.013
- **Love J**, Wimmer MT, Toth DJA, Chandran A, Makhija D, Cooper CK, Samore MH, Keegan LT (2021) Comparison of antigen- and RT-PCR-based testing strategies for detection of Sars-Cov-2 in two high-exposure settings. *PLoS ONE* 16(9):e0253407. doi:10.1371/journal.pone.0253407

- **Love J**, Keegan LT, Angulo FJ, McLaughlin J, Shea KM, Swerdlow DL, Samore MH, Toth DJA (2021) Continued need for non-pharmaceutical interventions after COVID-19 vaccination in long-term-care facilities. *Scientific Reports*. 11:18093. doi:10.1038/s41598-021-97612-w
- Goller F, **Love J**, Mindlin BG (2021) Song learning, frequency control and the exploitation of frequency space in passerines. *Ecology and Evolution*. doi:10.1002/ece3.7510
- **Love J** and Goller F (2021) Processes underlying complex patterns of song trait evolution in a *Setophaga* hybrid zone. *Ecology and Evolution*. doi:10.1002/ece3.7559
- **Love J** and Zelikowsky M (2020) Social density stress as a continuum. *Frontiers in Systems Neuroscience* 14. doi:10.3389/fnsys.2020.582985
- **Love J**, Hoepfner A, Goller F (2019) Song feature specific analysis of isolate song reveals interspecific variation in learned components. *Developmental Neurobiology* 79(4):350-369. doi:10.1002/dneu.22682

Published Conference Proceedings

- Nguyen DT, Duong KNC, Coates E, Nelson RE, **Love J**, Jones MM, Samore MH, Chaiyakunapruk N, Toth DJA P13 Assessment of the Effects of an Equity-Targeted Vaccination Outreach Program on Overall and Distributional Health and Economic Outcomes (2024) *Value in Health* 27(6):S3-S4. doi:10.1016/j.jval.2024.03.039
- Duong KNC, Nguyen DT, Kategeaw W, Liang X, Khaing W, Visnovsky LD, Veettil SK, McFarland MM, Nelson RE, Jones BE, Pavia AT, Coates E, Khader K, **Love J**, Vega Yon GG, Zhang Y, Willson T, Dorsan E, Toth DJA, Jones MM, Samore MH, Chaiyakunapruk N (2024) Incorporating social determinants of health into transmission modeling of COVID-19 vaccine in the US: a scoping review (2024) *Value in Health* 27(6):S175-S176. doi:10.1016/j.jval.2024.03.952
- Toth DJA, Sheets T, Beams A, Keegan L, **Love J**, Ahmed S, Zhang Y, Orleans B, Seegert N, Looney A, Alder S, Samore MH (2023) Novel Epidemiological Household Models Applied to Sars-Cov-2 Data from Over 15,000 Households in Utah: Insights into Transmission Variability and Age Effects. 9th international conference on infectious disease dynamics: o13.2

Popular Science Writing

- **Love J** (2015). View from the field. 2015 Annual Report of the Institute for Bird Populations.

In Prep

- Carroll E*, **Love J**, Kennedy A, Zelikowsky M (*in prep for Science*) Neural underpinnings of mating behavior and their dynamics under social isolation.

Research Funding

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- (2022) **PENDING/RESCINDED** (*subject to availability of funds*) CDC BAA 75D301-22-R-72097 “Investigating the role of vaccination in SARS-CoV-2 evolution” (**PI: Love**)
- Note: Recommended for funding by CDC (\$526,321.20 over 1 year), but funds have not been made available by US congress, so advice from CDC

is that this grant is in permanent limbo. Included here to illustrate a near-success in competitive research funding as PI while a research associate.

2022 CDC 1 U01 CK000675-01 “TRANSMIT: Training Research Acumen iN Students Modeling Infectious Threats” (PI: Keegan, **Co-I: Love**) (\$890k over 3 years)

Teaching

2024 Research stream in **Integrative Computational Biology**, Science Research Initiative, University of Utah (Instructor of Record)
 - Guided in-depth research experience for 5 students, 6 credit hours, 2 semesters, with lectures ranging from study design to genetics to complex systems. With guided research in social behavior evolution.

2023 **Modeling for Public Health Practitioners**, University of Utah Division of Epidemiology
 - Co-designed and delivered three 40 hour workshops of active learning material for this CDC-funded initiative to increase use of models by state and local public health administrations.

2021 Invited Guest Lecture: **Parasitology**, Hobart and William Smith Colleges.
 Lecture topic: *Basic Infectious Disease Epidemiology*

2014-2018 Teaching Assistant:
 - 7 sections **Evolution and Diversity of Life**: Independently prepared materials and lectured in weekly 1 hour discussion sections (~30 students), individual tutoring, etc.
 - 1 section **Ecology and Evolution**: Taught weekly 1 hour discussion sections (~30 students), individual tutoring, etc.
 - 1 section **Desert Ecology**: 10 day field course in Utah and Arizona. Helped 12 students develop research projects involving plant and avian communities in the Sonoran Desert. Taught students to identify species, take field notes, manage data, conduct basic analyses, and present their research in a poster and a paper.

Service

Research Advisor for:

- Kyah Hegreberg – 2024, SRI Freshman Undergraduate
- Beckett Chambers – 2024, SRI Freshman Undergraduate
- Dylan Chamas – 2024, SRI Freshman Undergraduate
- Tiffany Nguyen – 2024, SRI Freshman Undergraduate

Research Co-Advisor for:

- Andrea Chen – 2025-present, PhD in Computational Biology with Andrew Hein and Philipp Messer

- Cormac LaPrete – 2022-2024, MS in Applied Mathematics with Fred Adler (current: Program Specialist, Wasatch Adaptive Sports)
- Isaac Griss Salas – 2021, REU with Fred Adler (afterwards: MS [2022] and PhD [2025] in Applied Mathematics at U of Washington; current: Postdoc, U of Utah HCI)
- Erin Carroll – 2020, UROP with Moriel Zelikowsky (current: MD/PhD student, U of Utah)

Public health service for:

- Utah Department of Health and Human Services – 2024, forecasting and scenario analysis in support of the development of interventions to prevent future cases of congenital syphilis
- CDC – 2022, National Response to the 2022 Monkeypox Virus Outbreak, modeling team, 60 day emergency deployment under Ian Spicknall
- Utah Department of Health and Human Services – 2021, impromptu modeling support for public health decisionmaking relating to SARS-CoV-2

Reviewer for:

Wilson Journal of Ornithology; Clinical Infectious Diseases; Acta Zoologica Academiae Scientiarum Hungaricae (now Animal Taxonomy and Ecology); MIT Rapid Reviews: COVID-19; PLoS ONE; BMC Infectious Diseases

Other:

- Scientific consultant for an upcoming exhibition on collective behavior at the Museum of the Earth (opening 2027)

Fellowships, Scholarships, and Awards

2025	Gordon Research Conference, Travel Award
2021	University of Washington Biostatistics SISMID, Scholarship
2019	University of Utah, Riser Award for Outstanding Graduate Research
2018	University of Utah Dept. of Biology, Riser Research Fellowship American Ornithological Society, Travel Award
2017	University of Utah Dept. of Biology, Riser Research Fellowship
2016	National Science Foundation, GRFP, Honorable Mention University of Utah Dept. of Biology, George R. Riser Travel Award
2015	University of Utah Dept. of Biology, George R. Riser Travel Award
2014	AmeriCorps, Segal AmeriCorps Education Award
2007 – 2010	Georgia Student Finance Commission, HOPE Scholarship
2006 – 2010	University of Georgia, Berry College, Dean's List
2006	Berry College, Merit-based Scholarship

Selected Presentations and Outreach

Invited Research Presentations

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| 2026 | - <i>The collective context of predator-prey interactions and the evolution of collective movement</i> (invited talk) Gordon Research Conference on Predator-Prey Interactions: Integrating Theoretical Models and Big Data to Understand Predator-Prey Dynamics, Lucca, Italy, 1/27 |
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- *The evolution of collective behavior: how machine learning is unlocking a new scale of study* (invited seminar) Institute for Advanced Computational Science and Department of Ecology and Evolution, Stony Brook University, Stony Brook, NY, 7/9

Contributed Research Presentations

- 2026 - *The evolution of collective movement in fishes* (poster) Gordon Research Conference on Predator-Prey Interactions: Integrating Theoretical Models and Big Data to Understand Predator-Prey Dynamics, Lucca, Italy, 1/27
- *The evolution of collective movement in vertebrates* (poster) Gordon Research Conference on Unifying Ecology Across Scales: Connecting Unifying Ideas with Bold Data Streams to Predict Change in Natural Systems, Boston, MA, 7/15
- 2025 - *Investigating the evolution of collective movement in fishes* Evolution 2025, Athens, GA 6/21
- 2021 - *Comparison of antigen- and RT-PCR-based testing strategies for detection of Sars-Cov-2 in two high-exposure settings* University of Utah 3i COVID-19 Symposium, Virtual, 6/4
- 2019 - *Patterns of signal evolution reflect trait complexity in a Setophaga hybrid zone* Behavior 2019, Chicago, IL 7/24
- 2018 - *Song-feature specific analysis of isolate song reveals interspecific variation in learned components* (poster), International Ornithological Congress, Vancouver, BC, Canada 8/22
- *Analysis of isolate song reveals interspecific variation in learned components* American Ornithological Society meeting, Tucson, AZ, 4/11

Invited outreach

- 2022 - *Becoming a Guest Researcher at CDC* 2022 MInD Meeting, CDC, Atlanta, GA 9/12
- 2018 - *Using spectrographic visual representation to learn birdsong* IBP NCCN Seasonal Training, Friday Harbor, WA 5/25
- 2017 - *Understanding songs of hermit and Townsend's warblers and their hybrids* North Olympic Science Night, Port Angeles, WA 7/5
- 2016 - *Songs of hermit and Townsend's warblers on the Olympic Peninsula* IBP NCCN Seasonal Training, Astoria, OR, 5/30
- 2015 - *Convergent character displacement in an avian hybrid zone?* IBP NCCN Seasonal Training, Port Angeles, WA, 6/1

Selected Mentee Presentations

- 2025 - *Neural underpinnings of mating behavior and their dynamics under social isolation* Neuroscience 2025, San Diego, CA, 11/15 (Poster by co-mentee Erin Carroll)
- 2021 - *Modeling the spread of SARS-CoV-2 in healthcare settings* 3i COVID-19 Symposium, Virtual (Poster by co-mentee Cormac LaPrete)

Selected Workshop Attendance

2022	SISMID, University of Washington Biostatistics, Seattle, WA
2021	5 th Workshop on Viral Dynamics, Virtual SISMID, University of Washington Biostatistics, Seattle, WA
2020	Normative Modeling, COSYNE, Denver, CO
2018	Bioacoustics in R, American Ornithological Society meeting, Tucson, AZ

Non-academic Research Positions

2014-2018	Contract Biologist, Institute for Bird Populations
2011; 2013	Avian Field Technician, Institute for Bird Populations
2006	Biology Field Assistant, Berry College Longleaf Pine Project

Affiliations

American Society of Naturalists